

## Lec 01:-

### cs 218 Data Structures and Algorithms

- Python
- Course intro
  - Language choice
    - Language strength
    - Market
    - Ex-students
    - "C++ is dead"
  - Lab 3+1
  - Review

#### Week

#### Topic

week 1

Introduction to the course, why python, python refresher  
OOP in python, Fibonacci numbers and order of growth.

week 2

Linked list, Doubly linked list

week 3

Circular linked list  
Stack, stack case studies



week 4

Queue data structures with arrays  
Set data structure

week 5

Recursion, types of recursion, Tower of hanoi,  
fib revisited, summing a list

week 6

Sessional - I  
Introduction to Trees

week 7

Tree traversal  
Binary search tree

week 8

Heap  
Sorting introduction, how humans sort



week 9

Bubble sort, merge sort, insertion sort  
Quick sort, selection sort

week 10

Hashing and collision detection  
Greedy algorithms imp

week 11

Dynamic programming imp  
Session-II

week 12

Introduction to graph data structures  
Graph traversal

week 13

Finding shortest path in graphs

note

Graphs are  
the general  
DS to solve  
40% to 50%  
of problems.



week14

Metrices and vectors in python  
python collections module

week15

Standard library for OS traversals  
and other case studies

week16

Scaling dictionaries to real world  
Course review